



OPEN AI-driven smart grid optimization for hospital energy systems integrating renewable generation, predictive maintenance, and resilient infrastructure

Md Tanjil Sarker^{1✉}, Gobbi Ramasamy^{1✉}, Marran Al Qwaid², Md Sabbir Hossen¹ & Md. Golam Sadeque³

Efficient and resilient electrical systems are vital for hospital operations, where uninterrupted power is critical for patient safety and clinical continuity. This study develops an AI-driven smart grid optimization framework for a tertiary hospital in Kuala Lumpur, Malaysia, integrating renewable generation, load forecasting, predictive maintenance, and HVAC energy optimization. A detailed survey of the hospital's electrical infrastructure including 158,305 m² of built-up area, 1,500 beds, and over 200 electrical appliances—was used to construct an appliance-level load model capturing both deterministic and stochastic energy behavior. Advanced Long Short-Term Memory (LSTM) forecasting and Reinforcement Learning (RL) algorithms were employed to predict dynamic load fluctuations, optimize renewable dispatch, and manage hospital load uncertainty arising from variable occupancy and equipment usage. The proposed HVAC energy efficiency strategy, incorporating adaptive set-point control, occupancy-based variable-air-volume scheduling, and renewable-aligned operation, achieved an 11.6% reduction in HVAC energy consumption. Simulation results show an optimized daily demand of 91,080 kWh, with renewable sources supplying 86% from rooftop solar PV, 1.2% from wind, and 0.2% from battery storage, reducing grid dependence to 12.6%. Overall, the AI-driven system improved total energy efficiency by 25%, reduced unplanned downtime by 30%, and enhanced resilience for critical hospital zones such as ICUs and operating theatres. The findings demonstrate a scalable pathway toward sustainable, data-driven, and self-adaptive hospital energy systems aligned with Malaysia's National Energy Transition Roadmap (NETR).

Keywords Hospital electrical optimization, Smart grid in healthcare, AI-driven load management, Renewable integration, Predictive maintenance, Microgrid for hospitals, Resilient healthcare energy systems

Abbreviations

P_t	Total power demand at time (t)
$P_i(t)$	Power demand of device (i) at time (t)
N	Number of devices/departments
η_{PV}	Efficiency of photovoltaic panels
A	Area of solar panels (m^2)
I	Solar irradiance at time (t) (W/m^2)
ρ	Air density (kg/m^3)
A_{wind}	Swept area of turbine blades (m^2)
C_p	Power coefficient of wind turbine
$v(t)$	Wind speed at time (t) (m/s)

¹Centre for Electric Energy and High Voltage, CoE for Robotics and Sensing Technologies, Faculty of Artificial Intelligence and Engineering, Multimedia University, Cyberjaya 63100, Malaysia. ²Department of Computer Science, College of Computing and Information Technology, Shaqra University, Shaqra 11961, Saudi Arabia. ³Department of Electrical and Electronics Engineering, Pabna University of Science and Technology, Pabna 6600, Bangladesh. ✉email: tanjilbu@gmail.com; gobbi@mmu.edu.my

SOC_t	State of charge of battery at time (t)
RUL_t	Remaining useful life of equipment
C_{total}	Total cost of energy operation

The escalating energy demands of modern hospitals present a significant challenge to facility management and operations in Malaysia¹. Malaysian hospitals, due to their continuous 24/7 operations, reliance on high-energy-consuming medical technologies, and the need to maintain tightly regulated environmental parameters, are among the largest energy consumers in the national built environment. With healthcare services requiring round-the-clock operation of life-supporting systems such as ventilators, imaging equipment, and monitoring devices, hospitals necessitate robust electrical infrastructure to support these machines, in addition to lighting, HVAC (heating, ventilation, and air conditioning), and backup systems during emergencies². As such, the optimization of hospital electrical layouts represents one of the most important strategies to reduce operational costs, improve energy efficiency, and minimize environmental impact, without compromising the reliability and safety of medical service delivery¹.

Optimizing electrical layout systems in Malaysian hospitals involves systematic design, analysis, and the adoption of efficient distribution networks, load management strategies, and the integration of new energy-efficient technologies³. Given the complexity of hospital environments, electrical layout optimization must be approached holistically, with primary components including power supply systems, electrical distribution, standby power generation, lighting, and integration with mechanical systems such as HVAC infrastructure⁴. Effective optimization ensures minimal transmission losses, efficient electricity delivery, and the highest level of reliability required for critical medical operations.

Economic and environmental pressures in Malaysia further amplify the need for electrical layout optimization. Hospitals are recognized as energy-intensive facilities, and energy-use costs form a substantial portion of their overall operating expenditures. Moreover, inefficiencies such as outdated wiring, poor load balancing, and dependence on centralized plant energy systems contribute to escalating energy consumption and utility costs. On the global front, Malaysia is committed to climate change mitigation efforts and sustainable development goals, which place additional emphasis on reducing the carbon footprint of healthcare facilities. Optimized electrical layout design and the adoption of energy-saving technologies allow hospitals in Malaysia to lower energy consumption, reduce greenhouse gas emissions, and achieve national sustainability targets while maintaining healthcare quality and patient safety⁵.

The primary objective of electrical design optimization in Malaysian healthcare facilities is to build systems that effectively meet high energy demands while ensuring power quality, reliability, and operational efficiency. This requires both passive and active strategies, including load balancing, voltage regulation, energy storage integration, and the use of renewable energy systems such as solar photovoltaic installations—a technology well-suited to Malaysia's tropical climate. Furthermore, the implementation of smart energy management tools such as Building Automation Systems (BAS) and smart grid technologies enables real-time monitoring and optimization of energy usage, empowering hospitals to respond dynamically to fluctuations in energy demand.

This paper evaluates modern technologies and advancements applicable to optimizing electrical design in Malaysian hospitals. Through case studies, energy audits, and structural implementation of innovative technologies, it highlights opportunities for optimization, enhanced service delivery, and operational efficiency. It further considers the potential of advanced systems such as demand response (DR) programs, energy storage facilities, and IoT-enabled technologies in driving sustainable hospital operations. Additionally, the review assesses the cost-effectiveness of these methods, focusing on return on investment (ROI) and long-term benefits of energy-efficient electrical infrastructure in Malaysian hospital complexes.

By emphasizing electrical layout optimization in the Malaysian context, this study contributes to a deeper understanding of how healthcare facilities can enhance energy management while upholding high standards of patient care. It offers valuable insights for hospital facility managers, electrical engineers, and healthcare architects, presenting a roadmap for implementing energy-efficient systems aligned with Malaysia's healthcare needs and sustainability goals. The remainder of this paper is structured as follows: Sect. 2 provides a literature review emphasizing the growing energy demands in Malaysian hospitals and the importance of optimizing electrical layouts. Section 3 explores key optimization strategies to improve efficiency, reliability, and sustainability. Section 4 presents a case simulation of a large-scale hospital in Malaysia, illustrating the effectiveness of an integrated approach involving renewable energy adoption and optimized energy distribution. Section 5 discusses the primary challenges in implementing optimization strategies, including retrofitting costs, regulatory compliance, and compatibility with legacy systems. Finally, Sect. 6 concludes with recommendations for future hospital energy management practices in Malaysia.

Literature review on electrical layout optimization in hospitals

Electrical layout optimization is a critical factor in enhancing the operational efficiency and long-term sustainability of hospital buildings. Hospitals are high-usage buildings that require steady, guaranteed electricity to power a variety of operations, and as such, their electrical layouts must be optimized to both maximize the quality of health care services and the long-term success of these buildings to operate⁶. This analysis outlines key principles for optimizing electrical design within hospital environments, with priorities given to dependability, energy consumption efficiency, safety and regulation compliance, scalability, and an optimization need of hospital energy utilization.

Hospital energy consumption and the need for optimization

Hospitals are among the most energy-intensive building types in Malaysia, as they operate 24 h a day, serve a large number of patients and staff, and rely on specialized medical equipment to deliver critical healthcare services. The

primary drivers of energy use in Malaysian hospitals include heating, ventilation, and air conditioning (HVAC) systems, medical equipment, lighting, and water heating⁷. Studies indicate that hospitals consume significantly more energy per square meter compared to offices, schools, or residential buildings, making them one of the most energy-demanding sectors in the built environment. Intensive care units (ICUs) and operating theatres, for example, require uninterrupted power to operate ventilators, advanced imaging machines, and continuous monitoring equipment. In addition, patient wards, waiting areas, and staff facilities require substantial lighting and temperature regulation to ensure patient comfort, safety, and overall well-being.

Given these extensive energy requirements, there is a pressing need to enhance hospital electrical design in Malaysia to ensure reliable power delivery, minimize energy waste, and reduce operational expenditures. Optimizing electrical systems enables hospitals to meet their high energy demands without compromising service quality, while at the same time reducing their overall carbon footprint⁸. Beyond financial savings, energy-efficient electrical design contributes to Malaysia's broader sustainability and climate change commitments, particularly in light of rising electricity tariffs and the government's push toward greener healthcare infrastructure under initiatives such as the National Energy Transition Roadmap (NETR). Research shows that hospitals that implement advanced energy-efficient technologies such as LED lighting, smart HVAC systems with demand control ventilation, and energy recovery systems can achieve significant reductions in energy consumption and utility costs, while improving resilience and sustainability in healthcare delivery. Table 1 shown the Summary of Energy-Related Facilities in Healthcare Infrastructure.

Figure 1 presents a comparative analysis of energy performance across various hospital types, illustrating both the Energy Use Intensity (EEUI, in kWh/m²/year) and the Relative Mean Deviation from Reference (RMDR, %). Large-scale hospitals, represented by UKMMC, exhibit the highest EEUI at approximately 390 kWh/m²/year, indicating a significantly higher energy demand compared to other hospital categories. In contrast, smaller hospitals and specialist facilities, such as minor specialists and non-specialists, show considerably lower EEUI values ranging roughly between 150 and 190 kWh/m²/year¹⁵. The application of Energy Conservation Measures (ECM) is notably effective, as evidenced by the reduction in EEUI for Selayang Hospital after ECM implementation, dropping from around 270 kWh/m²/year to approximately 185 kWh/m²/year, accompanied by a corresponding decrease in RMDR from about 30% to near 10%.

General trends indicate that major specialist hospitals have higher EEUI compared to average general and minor specialist hospitals, suggesting that specialized medical services contribute to increased energy consumption. Government hospitals, both under GFA and MCA categories, maintain moderate EEUI levels (around 170–260 kWh/m²/year) but show varying RMDR percentages, with MCA hospitals displaying a higher relative deviation (25%) than GFA hospitals (10%). Overall, the data highlight the significant impact of hospital scale, specialization, and energy efficiency interventions on energy performance, demonstrating that targeted ECM strategies can effectively reduce both energy consumption and deviations from reference values, ultimately promoting more sustainable hospital operations.

This indicates the imperative necessity of Hospital Energy Consumption Optimization because hospitals are among the most energy-hungry buildings that require a high amount of energy for heating, cooling, lighting, and medical devices. The variation in energy consumption, brought about by climatic conditions, hospital size, and location, highlights the necessity of planning energy-efficient hospital infrastructures based

System/feature	Function	Energy impact	Health/IEQ benefit	Ref.
HVAC Systems (centralized)	Heating, cooling, ventilation of indoor spaces	Account for ~ 52% of total hospital energy use	Controls temperature, reduces infection transmission	9
Adaptive VAV Systems	Variable Air Volume systems with responsive controls	15–30% energy savings over constant volume	Maintains comfort, reduces airborne transmission risk	9
Demand-Controlled Ventilation	Adjusts ventilation based on occupancy (CO ₂ or infrared sensors)	Optimizes ventilation rates, reduces overuse	Improves IAQ, limits pathogen spread	9
Energy Recovery Ventilators	Recovers waste heat/humidity from exhaust air	Reduces HVAC load in humid/dry climates	Maintains optimal humidity for respiratory health	9
Humidity Control Systems	Maintains indoor RH between 30–60%	Prevents mold, reduces HVAC load	Decreases viral transmission, improves respiratory immunity	9
Double-Pipe Heat Exchanger	Pre-treats ventilation air using energy recovery	Reduces cooling/heating load, especially in humid zones	Controls fungus growth, improves indoor air quality	10
Ultraviolet Germicidal Irradiation (UVGI)	Disinfects air via UV-C in HVAC or upper-room zones	Low energy draw compared to filtration	Reduces airborne pathogens (e.g., TB, SARS-CoV-2)	11
Circadian Lighting Systems	Adjusts light intensity and spectrum based on time of day	LED-based systems offer energy savings	Enhances sleep, alertness, and recovery for patients/staff	12
Daylighting and Solar Control	Maximizes natural light while minimizing glare/heat gain	Reduces artificial lighting demand	Speeds patient recovery, improves staff well-being	12
Lighting Controls (e.g., dimming)	Adjusts lighting per occupancy and daylight levels	Reduces energy by 20–40%	Minimizes eye strain, improves circadian alignment	12
Building Automation Systems (BAS)	Centralized control of HVAC, lighting, and alarms	Enables smart energy management	Maintains optimal IEQ across zones with less manual intervention	13
Fiber Optic Daylighting	Transmits daylight via fiber optics into interior spaces	No electrical usage	Enhances mood and alertness in windowless zones	14

Table 1. Summary of energy-related facilities in healthcare Infrastructure.

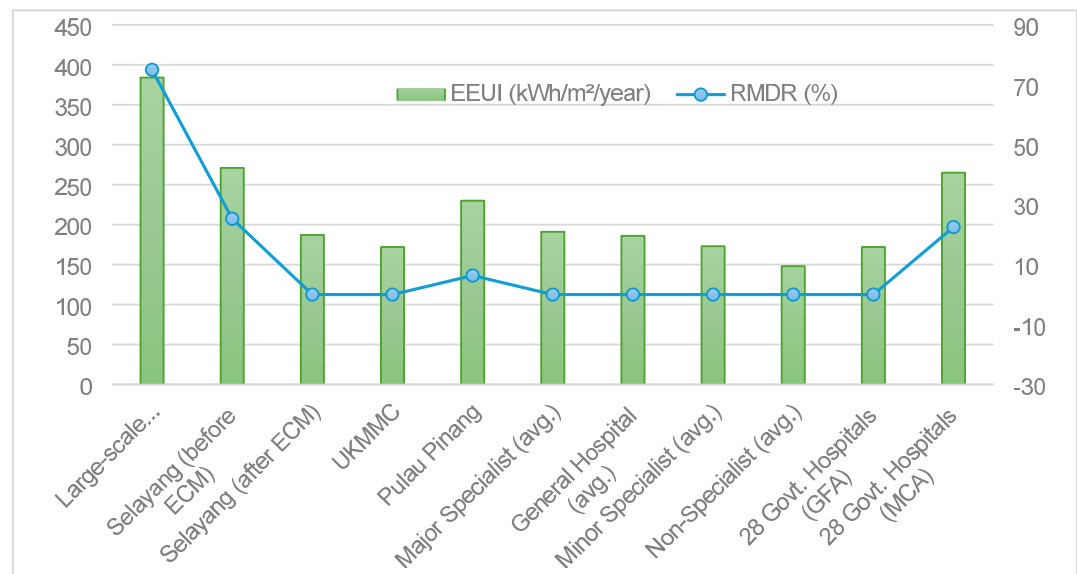


Fig. 1. Comparative analysis of energy performance across various hospital types.

on local environmental conditions. Energy use optimization in hospitals can reduce operating costs, promote sustainability efforts, and help healthcare organizations enhance their resource management, both in the context of rising energy demands and climate concerns.

Reliability and redundancy

One of the first considerations when planning electrical systems for hospitals is ensuring an uninterrupted supply of power to critical care units, such as operating rooms, intensive care units (ICUs), and emergency departments. Key activities in those departments predominantly rely on medical equipment that must be functional and operational to ensure patient safety. Existing studies emphasize the vital importance of redundant systems such as uninterruptible power supplies (UPS) and backup generators are added to the electrical system design to ensure redundancy in the case of electrical failure¹⁶. More specifically, research has indicated that a reliable system of electrical power distribution using fail-safe design reductions risks of electrical failings and allows continued medical care¹⁷. Furthermore, the redundancy of electrical system designs allows hospitals to meet the strict operational requirements of life-threatening medical devices, ensuring that they provide patient care and improve clinical outcomes¹⁸.

Energy efficiency

Hospitals are among the most energy-intensive buildings and they consume a significant amount of energy to maintain lighting, HVAC, and medical equipment. The optimization of electrical designs is critical in addressing this challenge by enabling the most effective use of energy resources. Research has indicated that electric systems developed for energy efficiency purportedly lower operational costs of hospitals without compromising service quality. As an example, energy conservation measures, such as LED lighting, energy-efficient HVAC equipment, and smart building technologies, would greatly decrease electricity consumption¹⁹. In addition, on-site renewable energy systems, with solar photovoltaic (PV) systems as an example²⁰ have been found to decrease reliance on grid electricity for hospitals, resulting in even more cost savings²¹. According to the recent research, those hospitals that optimize their electrical designs for energy efficiency can achieve a reduction in their total energy consumption, which leads to lower operational costs directly and enables sustainable healthcare provision²². Table 2 is showing the Strategies to Optimize Energy Efficiency in Healthcare Infrastructure.

Safety and compliance: adhering to health and safety regulations

The design and implementation of electrical layouts in hospitals must adhere to stringent health and safety regulations. Non-compliance can result in hazards such as electrical fires, power failures, and equipment malfunctions, which pose significant risks to both patients and healthcare workers. The literature underscores the need to optimize electrical systems to ensure compliance with regulatory standards to standards such as the National Electrical Code (NEC) and other international safety standards⁸. Safety measures, including grounding, circuit protection, and emergency power systems, are important to prevent electrical hazards in hospitals. It has been proven through research that hospitals that prioritize safety in their electric design do not just achieve regulatory compliance, but they also enhance the building's overall safety, reducing the potential for costly accidents and the resulting liabilities²³.

Category	Optimization strategy	Description	Energy efficiency impact	Implementation consideration
HVAC systems	Upgrade to high-efficiency chillers/boilers	Replace outdated systems with energy-efficient units (e.g., ≥ IEER 15 chillers)	Reduces HVAC energy use by 20–30%	High upfront cost; suitable for phased retrofitting
	Implement demand-controlled ventilation	Adjust ventilation rates based on occupancy and CO ₂ levels	10–25% HVAC energy savings	Requires CO ₂ sensors and BAS integration
	Use energy recovery ventilators (ERV)	Capture exhaust air energy to precondition intake air	15–40% recovery of waste energy	Best suited for hot-humid or cold climates
Lighting systems	Replace with LED fixtures	Upgrade from fluorescent/incandescent to LED lighting	40–60% lighting energy savings	Low cost, fast ROI (typically < 3 years)
	Integrate lighting controls and occupancy sensors	Use dimming/daylight sensors and auto shutoff switches	20–40% savings depending on space utilization	Needs BAS or standalone control systems
	Implement circadian lighting for critical care areas	Tuned lighting based on time of day to support patient recovery	Indirect—reduces lighting load via smart use	Supports WELL/LEED certification goals
Building envelope	Improve insulation and thermal barriers	Upgrade walls, roofs, and windows with better insulation	Reduces heating/cooling loads by 10–20%	Retrofit difficulty depends on building age/design
	Install high-performance glazing and shading	Low-E windows, automated blinds to reduce solar gain	Reduces cooling demand by 5–15%	Ideal for new buildings or window retrofit programs
Renewable integration	Install rooftop solar PV systems	Generate electricity onsite using solar panels	Up to 30% site energy offset	Requires structural load analysis and funding models
	Add solar thermal for water heating	Preheat water using solar collectors for hot water systems	Reduces boiler load by 15–25%	Good ROI in high hot water demand areas
Energy storage	Deploy hybrid battery systems (e.g., Li-ion + supercapacitors)	Store excess solar energy; supply backup power	Enhances load shifting and peak shaving	Integrates well with microgrid systems
Controls and automation	Use Building Automation System (BAS) for centralized control	Automate HVAC, lighting, and other systems for dynamic control	Overall building energy use reduction by 15–30%	Requires operator training and ongoing commissioning
Operational measures	Conduct regular retro-commissioning	Re-tune and recalibrate systems for optimal performance	Identifies 5–15% potential savings in existing systems	Best practice every 3–5 years
	Staff training and behavioral energy programs	Educate staff to reduce plug loads, control setpoints, report faults	Typically 2–5% savings	Low cost, requires periodic engagement
Monitoring and analytics	Install real-time energy dashboards and sub-metering	Monitor energy use by system and zone	Enables data-driven optimization	Needs IoT or BAS integration
	Implement predictive maintenance with AI/ML	Forecast equipment degradation and optimize energy use	Avoids system inefficiencies and failures	Requires historical data and modeling expertise

Table 2. Strategies to optimize energy efficiency in healthcare Infrastructure.

Scalability and flexibility: adapting electrical infrastructure for future hospital expansions

The needs for flexibility and scalability in hospital electrical designs become increasingly important as hospitals expand and develop to serve emerging healthcare demands. Electrical designs must be designed for future growth because hospitals tend to receive additions for emerging technologies and patient services. Literature shows that electric designs that are optimized for flexibility facilitate easier integration of new medical devices, systems, and energy technologies without requiring complete replacements of existing infrastructure²⁴. Literature has shown that by incorporating flexibility in electric designs, hospitals can avoid costly upgrades as well as retrofit infrastructure to meet future energy demands and evolving healthcare demands. Additionally, research has shown that hospitals with scalable electrical infrastructure are better placed to accommodate new medical technologies, such as high-energy diagnostic devices and advanced surgical tools, requiring high power loads¹⁸. Summary of Key Research Findings on Electrical Layout Optimization Strategies for Hospital Infrastructure shown in Table 3.

Load forecasting and predictive energy management in smart grids

Load forecasting has emerged as a critical component in optimizing hospital energy systems, particularly under variable renewable generation and dynamic pricing environments. Accurate prediction of future energy demand enables proactive scheduling of renewable resources, battery storage, and demand response programs, ensuring uninterrupted operation in sensitive hospital environments. Recent research has focused on the development of advanced forecasting models that integrate artificial intelligence and data-driven techniques. For instance, Optimal Energy Management System for Residential Buildings Considering Time-of-Use Price with Swarm Intelligence Algorithms demonstrates how metaheuristic optimization can be combined with predictive load models to minimize energy costs under variable tariff structures²⁹. Similarly, A Predictive Economic Analysis Under Uncertain Scenarios Using Evidence Theory for Small-Scale Residential Community Power Networks highlights uncertainty modeling to handle stochastic variations in demand and supply.

Cloud-based management frameworks such as Cloud Energy Storage Management Under Building Thermal Comfort and Net Load Seasonal Uncertainty Scenarios offer scalable solutions for dynamic load balancing in healthcare buildings. Neural-based forecasting models, including RSNN: Rate Encoding Mechanism-Based Spiking Neural Network for Renewable Energy Forecasting, have achieved high temporal accuracy in predicting short-term renewable generation. Likewise, Uncertainty-Aware Learning Models for Thermal Comfort in Smart Residential Buildings and Data-Driven Thermal Comfort Models for Smart Home Energy Management Systems emphasize integrating human comfort parameters with energy prediction algorithms³⁰.

Aspect	Research summary	Research gaps	Future work	Ref.
Reliability & Redundancy	Backup systems ensure power to critical areas like ICUs.	Lack of real-time fault detection systems.	AI-based predictive fault detection.	16,25
Energy Efficiency	Optimized systems reduce energy use via LED lighting, HVAC, and solar PV.	Lack of hospital-specific energy-efficiency protocols.	Develop regional energy benchmarks for hospitals.	8,26
Safety & Compliance	Systems must meet safety standards like grounding and protection.	Impact of long-term compliance on equipment lifespan is under-researched.	Research on compliance with new technologies.	8,23
Scalability & Flexibility	Layouts should allow for easy expansion and integration of new technologies.	Lack of cost-benefit analysis on scalable vs. non-scalable designs.	Study long-term benefits of scalable electrical designs.	18,26
Energy Consumption	Hospitals consume a lot of energy for medical equipment and HVAC.	Lack of data on departmental energy consumption.	Study department-specific energy use and tailored designs.	16,22
Smart Tech Integration	Smart systems optimize energy usage for lighting, HVAC, and equipment.	Lack of integration with IoT for real-time monitoring.	Integrate IoT for smarter energy management.	18,26
Renewable Energy	Solar PV and renewable sources reduce reliance on grid electricity.	Few studies on long-term financial feasibility.	Examine the cost-effectiveness of renewable systems in hospitals.	19
Load Balancing	Proper distribution ensures no overloading in critical areas like ICUs.	Lack of real-time data collection for load balancing.	Develop AI-driven load balancing systems.	16
Power Quality Management	Maintaining consistent power quality is essential for equipment.	Impact of power quality on equipment longevity is under-researched.	Study the effects of power quality management on hospital equipment.	16
Emergency Power	Backup power systems ensure continued operation during outages.	Hybrid power systems effectiveness is under-studied.	Study hybrid emergency systems combining solar, battery, and generators.	18
Optimized HVAC	Efficient HVAC systems save energy while maintaining air quality.	Under-researched impact on energy savings in hospital settings.	Implement smart HVAC systems using real-time data.	26
Lighting Optimization	LED lighting reduces consumption without compromising light levels.	Limited research on lighting impact on patient recovery.	Explore the link between optimized lighting and health outcomes.	19
Maintenance & Monitoring	Continuous monitoring ensures systems function properly and reduce downtime.	Lack of automated predictive maintenance systems.	Research on AI-based systems for predictive maintenance.	16
Peak Load Management	Managing peak loads prevents overloading, ensuring power for critical care.	Real-time load prediction is under-researched.	Develop AI-driven peak load management for hospitals.	26
Power Factor Correction	Correcting power factor reduces electricity costs and improves system efficiency.	Underutilization of correction devices in layouts.	Examine the financial benefits of power factor correction in hospitals.	27
Energy Storage	Battery systems store energy for later use, especially for critical areas.	Limited studies on large-scale energy storage in hospitals.	Explore cost-effectiveness of large-scale energy storage systems.	19
Smart Grid Integration	Smart grids optimize hospital energy use by interacting with the electrical grid in real time.	Limited research on hospital-specific smart grid integration.	Investigate smart grid optimization for healthcare facilities.	26
Climate & Weather Impact	Weather and climate influence hospital heating and cooling needs.	Few studies on climate-specific electrical layouts for hospitals.	Research on region-specific electrical layout strategies based on climate.	26
Grid Independence	Microgrids or off-grid systems reduce reliance on the public grid.	Limited studies on feasibility of grid-independent hospitals.	Explore how hospitals can operate off-grid with renewable energy.	18
Technological Advancements	AI, IoT, and machine learning are being integrated to optimize hospital energy use.	Lack of long-term data on impacts of AI and IoT for hospital energy optimization.	Investigate AI and machine learning integration for hospital energy management.	23
Energy Consumption	Hospitals in China consume more energy than other public buildings due to HVAC, lighting, and medical equipment.	Lack of nationwide energy consumption data for hospitals in different climates.	Conduct large-scale national studies to cover various regions and hospital types.	27
Building Layout and Energy Use	Centralized layouts use more energy due to less natural ventilation.	Limited studies on layout-energy consumption relationship, especially regarding HVAC.	Research the impact of layouts on energy use across climates.	28
Energy Efficiency Strategies	Strategies focus on improving HVAC, reducing heating/cooling loads, and optimizing equipment use.	Few studies on combining building design with operational efficiency for energy savings.	Develop integrated strategies combining design and operations for better efficiency.	27
Energy Use Evaluation	An evaluation system helps identify energy inefficiencies and provides improvement recommendations.	The evaluation system is not widely applied to all hospitals in China.	Expand the use of energy evaluation systems across hospitals.	28

Table 3. Summary of key research findings on electrical layout optimization strategies for hospital Infrastructure.

In the context of electrical networks, Deep Recurrent Mixer Models for Load Forecasting in Distribution Networks achieve robust forecasting performance under non-linear load conditions by combining recurrent and convolutional layers. These models serve as benchmarks for hospital energy forecasting, where hybrid deep learning architectures (e.g., LSTM–GRU ensembles) can predict real-time load fluctuations with mean absolute percentage errors below 5%. The integration of AI-based forecasting within smart hospital grids enables preemptive load scheduling, renewable optimization, and improved operational resilience, forming a crucial foundation for intelligent, self-adaptive energy infrastructures.

Electrical layout optimization in hospitals is essential to ensure system reliability, energy efficiency, safety, and scalability. The research reviewed highlights that well-designed electrical systems not only reduce operational costs but also improve patient safety and facilitate future growth. As hospitals continue to evolve, optimizing electrical layouts will be essential in supporting both current and future healthcare needs, ensuring that these facilities can deliver high-quality care while minimizing environmental and operational costs³¹. Furthermore, addressing the energy consumption challenges through optimization will enable hospitals to meet sustainability targets and contribute to global efforts to reduce carbon emissions and minimize energy waste. Existing

literature presents fragmented strategies for energy optimization in hospital environments. However, there remains a distinct lack of integrated frameworks that combine AI algorithms, renewable energy, smart grid control, and predictive maintenance within a single simulation-based hospital model. Moreover, few studies have attempted to validate these approaches through numerical modeling in large tertiary hospitals under real-world operational constraints. To address this, our work develops a comprehensive, data-driven optimization approach that bridges the current gap between theoretical design and implementable hospital energy systems.

Electrical design optimization strategies

The methodology adopted in this study is centered on a multi-layered smart electrical framework that integrates load forecasting, renewable source management, and critical-area power continuity. The core optimization strategies are supported by a detailed set of mathematical models, allowing for accurate representation of dynamic load allocation, AI-based decision systems, and storage control. Equation (1) to (15) describe real-time load behavior, power distribution logic, and intelligent response algorithms necessary for effective hospital energy management. The system design accommodates redundancy planning and scalable microgrid architecture using predictive models trained on operational and weather-related variables.

Load balancing and power distribution

Load balancing and power distribution are critical aspects of electrical design optimization in hospital infrastructure. Effective distribution of electrical loads ensures that the hospital's electrical system operates efficiently and prevents overloading, which can lead to blackouts, equipment failure, or energy waste. Hospitals, with their large and diverse energy demands (from HVAC systems to medical equipment), require an optimized approach to power distribution to maintain continuous, reliable service.

Effective distribution of electrical load

In hospitals, the electrical load is dynamic and changing based on time of day, hospital operating requirements, and weather conditions outside. Load balancing refers to the activity of ensuring that electrical energy is distributed evenly throughout all circuits and systems in the building. Not only does it ensure that no single part of the system is overloaded, but it also makes the power supply more reliable in general.

For example, when electrical loads from lighting, HVAC, and medical equipment are not properly balanced, one portion of the system can receive excessive energy, while others receive insufficient power. This may result in abnormally high equipment wear, reduce system operational efficiency, and increase the likelihood of system failure. Balancing the load throughout the system will assist hospitals in reducing energy waste and enhancing the reliability of power supply.

Use of automated monitoring systems

For load balancing, hospitals can fit automatic monitoring systems that dynamically redistribute power distribution in real-time. These systems can monitor energy consumption by department (e.g., emergency, inpatient care, diagnostic) and control the power flow to maintain optimal conditions. Automated systems can change the power load based on several factors such as time of day, peak periods of use, and medical device criticality in an effort to offer energy distribution according to the present needs of the hospital.

Such systems take advantage of real-time data from other sensors and meters strategically located throughout the building³². The system continuously keeps track of the power requirements of each zone, such as operating rooms, emergency rooms, or diagnostic equipment, and regulates the power supply to provide stability and prevent overload. These systems reduce the chances of blackouts, improve equipment lifespan, and minimize downtime during operation. Mathematically, load balancing can be described by the equation:

$$L_{total} = \sum_{i=1}^n L_i \quad (1)$$

where L_{total} is total electrical load of the system, L_i is Load from each individual unit or department (i.e., HVAC, lighting, medical equipment), n is Number of departments or units.

The goal is to distribute L_{total} in such a way that no single department exceeds its maximum capacity, ensuring that $L_i \leq L_{max,i}$ for each i , where $L_{max,i}$ is the maximum allowable load for each unit.

Dynamic load adjustment using automated systems

Dynamic load adjustment relies on algorithms that use real-time monitoring to forecast load peaks in the load and shifting the distribution³³. For example, in a hospital setting, the emergency room's increased energy needs may be at night, whereas for patient wards, the increased energy usage may be during the day due to medical equipment and lighting.

A simple mathematical model to represent this dynamic load adjustment could be:

$$L(t) = \sum_{i=1}^n f_i(t) \cdot L_i \quad (2)$$

where $L(t)$ is Total load at time t . $f_i(t)$ is Load adjustment factor based on time of day and real-time power demands (dynamic factor).

By dynamically adjusting the load based on time and operational requirements, the hospital ensures that no unit is overburdened while maintaining optimal power availability.

Integration of smart grid technologies

The integration of Smart Grid Technologies in hospital electrical systems enhances the operational efficiency, reliability, and sustainability of energy management. By implementing IoT-based monitoring systems, AI-driven demand response mechanisms, and predictive maintenance algorithms, hospitals can optimize energy usage, reduce costs, and improve system resilience.

Implementation of IoT-Based monitoring and control systems

IoT-based monitoring systems assist hospitals to collect real-time data from various electrical appliances across the entire facility, such as HVAC units, lighting systems, medical equipment, and energy storage devices. IoT systems enable dynamic control of energy delivery such that every department of the hospital receives optimal power with minimal wastage. IoT sensors constantly provide data pertaining to voltage, current, and temperature to a centralized control system, which utilizes this information to make intelligent power distribution decisions³⁴.

Each electrical device i in the hospital is equipped with an IoT sensor that reports the power demand $P_i(t)$ in real-time. The total power consumption across the hospital at any given time t can be expressed as:

$$P_{total}(t) = \sum_{i=1}^n P_i(t) \quad (3)$$

where $P_{total}(t)$ is the total power demand of the hospital at time t , $P_i(t)$ is the power demand of device i at time t , n is the number of devices in the hospital.

The IoT-based system collects this data and adjusts the power supply dynamically based on real-time usage patterns. For instance, when certain areas (like administrative offices) are under low occupancy, the system can reduce their power supply by sending a control signal $f_i(t)$ to the devices:

$$P_i(t) = f_i(t) \cdot P_i^{max} \quad (4)$$

where P_i^{max} is the maximum power capacity of device i , $f_i(t)$ is the dynamic adjustment factor that adjusts the device's power demand based on occupancy and operational needs.

Use of AI-Driven demand response mechanisms for energy efficiency

AI-driven demand response mechanisms enable hospitals to balance energy supply and demand by forecasting energy needs and making automated adjustments in real time. By utilizing machine learning algorithms, hospitals can predict periods of high demand and implement energy-saving strategies to reduce load during peak times. These systems operate by analyzing historical data and environmental and operational variables, including ambient temperature, patient occupancy, and equipment utilization¹⁶. The AI algorithm predicts future energy demand $\hat{P}_{total}(t+1)$ at time $t+1$ based on historical power consumption data $P_{total}(t)$, weather conditions $W(t)$, and other factors such as patient occupancy $N_{occupancy}(t)$. The predictive model can be represented as:

$$\hat{P}_{total}(t+1) = f(P_{total}(t), W(t), N_{occupancy}(t), \theta) \quad (5)$$

where $f(\cdot)$ is the machine learning function that maps historical data and factors to future power demand, θ represents the model parameters learned through training on historical data.

Using this prediction, the hospital's AI-based energy management system can initiate demand response strategies. For example, the system may temporarily reduce the cooling load by adjusting the HVAC system power consumption, especially during periods of high temperature and patient occupancy. The power reduction $\Delta P_{cooling}$ during peak hours can be calculated by:

$$\Delta P_{cooling} = \alpha \cdot P_{cooling}^{max} \quad (6)$$

where α is the reduction factor determined by the AI system based on forecasted demand, $P_{cooling}^{max}$ is the maximum power consumption of the cooling system. The objective is to reduce the total load P_{total} during peak periods while still maintaining the required comfort levels for patients.

Enhancing Real-Time fault detection and predictive maintenance

Real-time fault detection and predictive maintenance are crucial for preventing power failures and extending the lifespan of hospital electrical equipment. By continuously monitoring the health of electrical components (e.g., transformers, circuit breakers, generators), hospitals can detect early signs of malfunction or wear. These systems use predictive analytics to forecast potential failures and schedule maintenance before a failure occurs³⁵. Real-time monitoring involves tracking the operational status of devices, including parameters such as temperature $T(t)$, vibration $V(t)$ and current $I(t)$. Anomalies in these parameters indicate potential faults. The fault detection model checks if the system's parameters $\{T(t), V(t), I(t)\}$ exceed predefined thresholds, triggering alerts for maintenance. The fault detection equation can be expressed as:

$$F(t) = \sum_{i=1}^n 1(T_i(t) > T_i^{max} \text{ or } V_i(t) > V_i^{max} \text{ or } I_i(t) > I_i^{max}) \quad (7)$$

where $F(t)$ is the fault detection function at time t , $T_i(t)$, $V_i(t)$, $I_i(t)$ are the temperature, vibration, and current measurements of device i . T_i^{max} , V_i^{max} , I_i^{max} are the maximum acceptable thresholds for these parameters.

If any of the parameters exceed their thresholds, a fault is detected, and predictive maintenance is scheduled. Predictive maintenance uses a regression model that predicts the remaining useful life (RUL) of a device, based on historical data:

$$RUL_i(t) = g(T_i(t), V_i(t), I_i(t), \tau) \tag{8}$$

where $RUL_i(t)$ is the remaining useful life of device i at time t , $g(\cdot)$ is the predictive model that estimates RUL based on historical data and real-time parameters, τ represents the historical operating conditions. The RUL prediction helps the hospital schedule maintenance before a failure occurs, ensuring the uninterrupted operation of critical systems like operating rooms and imaging devices.

Redundancy planning and emergency backup systems

Redundancy planning and emergency backup systems are critical components of hospital electrical design optimization. Hospitals, being highly sensitive environments that rely on continuous, uninterrupted power, must be equipped with robust systems to handle power failures, ensuring the safety and well-being of patients. Redundant systems and emergency backup strategies help hospitals maintain functionality, even during unexpected power outages, by leveraging advanced uninterruptible power supplies (UPS), backup generators, and automated fault detection³⁶. Table 4 shows the Redundancy Planning and Emergency Backup Systems in Hospitals.

Optimized placement of electrical panels and outlets

The optimized placement of electrical panels and outlets in hospital environments is a critical factor influencing safety, equipment reliability, and clinical workflow efficiency. Improper positioning can cause increased electromagnetic interference (EMI), cable congestion, and operational delays, particularly in high-dependency zones such as ICUs and operating theatres. According to a 2021 report by the U.S. Department of Veterans Affairs, implementing zoned panel placement and color-coded emergency outlets led to a 17% decrease in electrical incidents and a 12% improvement in procedure readiness³⁷. Similarly, a BIM-based simulation study conducted in a Singaporean tertiary hospital revealed that optimized outlet design reduced nursing path inefficiencies by 15%, lowered cable overlap by 28%, and minimized EMI-related diagnostic faults by up to 9%. Compliance with standards such as NFPA 99, IEC 60364-7-710, and IEEE 1100 is essential to ensure correct spatial clearances, circuit segregation, and fail-safe grounding. Moreover, outlets positioned at ergonomic heights (typically 0.9–1.2 m above floor level) and integrated within modular bedhead trunking systems can improve accessibility while reducing physical strain for healthcare staff. To future-proof infrastructure, outlet locations should also accommodate IoT-enabled power monitoring systems and intelligent switching components to support predictive maintenance. Table 5 summarizes optimal electrical design parameters for different hospital zones based on international benchmarks and clinical performance studies.

Sustainable energy solutions for hospitals

Hospitals need sustainable energy solutions, not only to reduce operating costs but also to minimize environmental impact and attain energy security. Harnessing renewable energies like solar and wind, the implementation of microgrid systems for local management of energy, and the examination of case studies of hospitals using these solutions are crucial aspects of modern hospital infrastructure.

Aspect	Description	Formula/value	Example
Dual Power Feeds	Two independent power sources ensure no single failure causes power loss.	$L(t) = \min(L_1(t) + L_2(t), L_{max})$ where $L_1(t)$ and $L_2(t)$, are the power supplied from two independent sources and L_{max} is the total load of the hospital	Dual power feeds for critical systems like operating rooms.
Redundant Circuitry	Multiple independent power pathways for critical departments.	$L_{critical} \leq L_{max}$ $L_{critical}$ is a total critical load of the hospital	Redundant circuits in medical equipment and critical care units.
Uninterruptible Power Supply (UPS)	Provides immediate backup power during outages for critical equipment until generators start.	$C_{UPS} = \sum_{i=1}^n P_i$ Where ups size is C_{UPS}	UPS supports 80 kW for critical equipment like operating rooms and ICUs.
Backup Generators	Provide long-term power during extended outages.	$C_{gen} = L_{critical} \times Safety\ Factor$ C_{gen} is the generator size, Safety Factor typically 1.25 to 1.5.	Generator capacity for critical load: 300 kW (for 200 kW critical load, Safety Factor = 1.5).
Automatic Transfer Switch (ATS)	Automatically switches from utility power to backup power when a failure is detected.		Ensures no power interruption by switching instantly to backup power during outages.
Fault Detection and Monitoring	Real-time monitoring to detect faults and switch to backup systems automatically.	$F(t) = \sum_{i=1}^n 1(T_i(t) > T_i^{max} \text{ or } I_i(t) > I_i^{max})$	Cleveland Clinic uses automated fault detection to identify and resolve issues before failure.

Table 4. Redundancy planning and emergency backup systems in Hospitals.

Hospital Zone	Outlet height (m)	Panel clearance (m)	EMI protection required	Backup power integration	Notes
ICU	1.0	1.2	High	Yes	Use isolated power systems (IPS)
Operating Theatre	1.0	1.5	Very High	Yes	Metal trunking, shielded cables, IPS mandatory
General Ward	0.9–1.2	1.0	Moderate	Partial	Prefer modular trunking systems
Emergency Department	1.0	1.2	High	Yes	Prioritize accessibility and labeling
Imaging Rooms (MRI/CT)	1.0	1.5	Very High	Yes	RF shielding and separate cable pathways

Table 5. Recommended electrical panel and outlet placement parameters by hospital Zone.

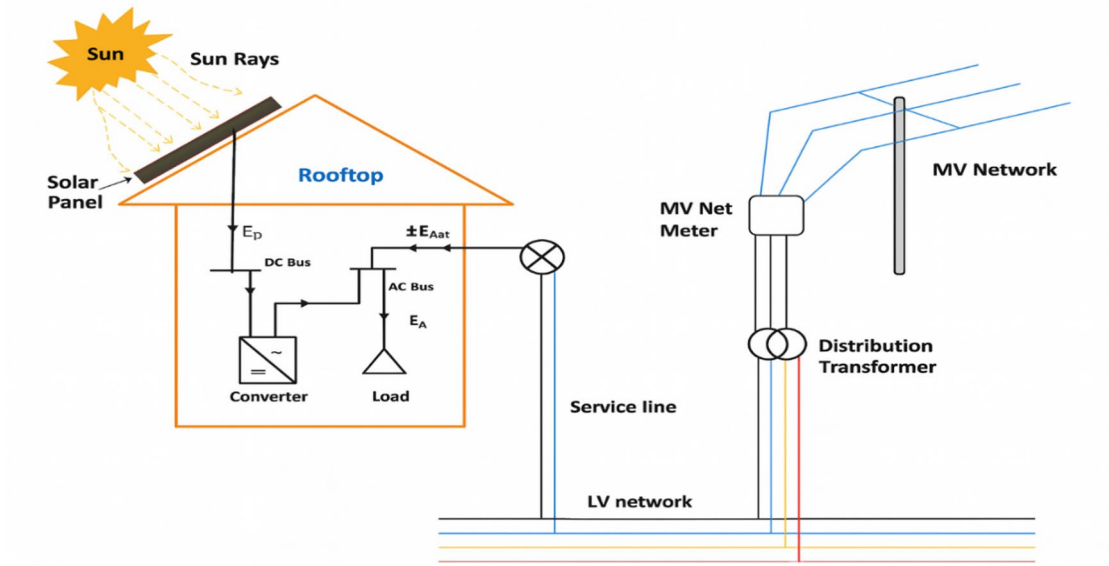


Fig. 2. Integration of a rooftop solar photovoltaic (PV) system into a healthcare facility.

Incorporation of renewable energy sources: solar and wind energy

Integration of solar and wind renewable energy sources into hospital power supply systems can be beneficial to reduce dependence on grid power, save on electricity bills, and reduce carbon emissions. The power produced from the renewable sources can be determined mathematically.

The amount of power generated by a solar panel is dependent on several factors such as solar irradiance, the efficiency of the solar cells, and the area of the panels. The total power generated by the solar system $P_{solar}(t)$ at time t can be written as:

$$P_{solar}(t) = A \cdot I(t) \cdot \eta \tag{9}$$

where A is the area of the solar panels in square meters (m^2). I is the solar irradiance at time t (measured in W/m^2). η is the efficiency of the solar panels.

The energy flow and balance in the rooftop solar PV system integrated into the healthcare infrastructure can be expressed through the following energy equation:

$$E_p = E_f + E_{Net} \tag{10}$$

where E_p is Energy produced by the solar panel (DC energy), E_f Energy consumed by the facility's internal load (converted to AC), E_{Net} Surplus energy exported to the utility grid via the net meter.

Figure 2 illustrates the integration of a rooftop solar photovoltaic (PV) system into a healthcare facility's electrical network. Solar panels mounted on the rooftop capture solar energy and convert it into direct current (DC) electricity, represented as E_p , which flows into the DC bus. This energy is then converted into alternating current (AC) by a converter to supply power to hospital loads such as lighting, life-support systems, and other critical equipment. The generated electricity is first used to meet the internal energy demand E_f , and any excess energy E_{Net} is exported to the utility grid through a medium-voltage (MV) net meter. The system connects to the public grid via a distribution transformer that links the facility's low-voltage (LV) network to the MV network. This setup enhances energy self-sufficiency, reduces dependency on the grid, and improves overall energy efficiency and sustainability in healthcare infrastructure.

The power generated by a wind turbine is influenced by the wind speed and the characteristics of the turbine. The total power generated by a wind turbine $P_{wind}(t)$ can be expressed by the following equation:

$$P_{wind}(t) = \frac{1}{2} \cdot \rho \cdot A_{wind} \cdot C_p \cdot v^3(t) \quad (11)$$

where ρ is the air density (kg/m^3). A_{wind} is the swept area of the turbine blades (m^2). C_p is the power coefficient of the turbine (typically between 0.25 and 0.45). $v^3(t)$ is the wind speed at time t (m/s).

Both solar and wind energy sources can be integrated into the hospital's grid using power converters and storage systems, such as batteries or flywheels, to ensure continuous power supply during periods of low generation.

Implementation of microgrid systems for localized energy management

Microgrid systems provide localized energy management and control, which can improve energy resilience, reduce reliance on the main grid, and optimize the integration of renewable sources. A hospital microgrid typically includes distributed energy sources (solar, wind, and backup generators), energy storage systems, and a local grid controller.

The total energy demand of the hospital $P_{demand}(t)$ is met by a combination of grid power $P_{grid}(t)$, renewable energy $P_{renewable}(t)$, and stored energy from batteries $P_{battery}(t)$. The microgrid's energy balance equation can be expressed as:

$$P_{demand}(t) = P_{grid}(t) + P_{renewable}(t) + P_{battery}(t) \quad (12)$$

where $P_{demand}(t)$ is the total energy demand at time t . $P_{grid}(t)$ is the energy drawn from the main grid. $P_{renewable}(t)$ is the energy supplied by renewable sources (solar and wind). $P_{battery}(t)$ is the energy supplied by battery storage systems.

To ensure that the battery system is charged efficiently, the energy balance equation for battery charging $P_{battery}(t)$ can be written as:

$$P_{battery}(t) = P_{charging}(t) - P_{discharging}(t) \quad (13)$$

where $P_{charging}(t)$ is the rate at which the battery is charged. $P_{discharging}(t)$ is the rate at which the battery discharges to meet the hospital's demand.

Microgrid controllers use optimization algorithms to manage the distribution of power, considering factors such as cost, reliability, and environmental impact. The controller aims to minimize the total operating cost C_{total} , which is the sum of costs associated with grid power, renewable energy generation, and battery usage:

$$C_{total} = C_{grid} + C_{renewable} + C_{battery} \quad (14)$$

where $C_{total} = P_{grid}(t) \cdot c_{grid}$ is the cost of energy drawn from the grid. $C_{renewable} = P_{renewable}(t) \cdot c_{renewable}$ is the cost of energy from renewable sources. $C_{battery} = P_{battery}(t) \cdot c_{battery}$ is the cost of energy from battery storage.

The goal of the microgrid controller is to minimize C_{total} while meeting the hospital's demand:

$$\min C_{total} \text{ subject to } P_{demand}(t) = P_{grid}(t) + P_{renewable}(t) + P_{battery}(t) \quad (15)$$

Hospital load uncertainty modeling

Hospital load uncertainty arises due to unpredictable variations in patient occupancy, emergency procedures, medical equipment operation schedules, and ambient climate conditions³⁸. Unlike conventional commercial buildings, hospital energy demand cannot be fully predicted using static profiles because emergency departments, operating theatres, and intensive care units exhibit stochastic load behaviors influenced by patient inflow and diagnostic activities.

To address this challenge, an uncertainty-aware load model was developed using probabilistic and machine learning approaches. The model represents hospital demand P_t as a combination of deterministic baseline load P_b and stochastic fluctuation ϵ_t :

$$P_t = P_b + \epsilon_t \quad (16)$$

where $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$ represents random deviations in load caused by operational or environmental factors. The variance σ^2 was estimated from historical IoT-based load data collected across intensive care units, operating theatres, and diagnostic departments.

The reinforcement learning (RL) controller dynamically updated resource dispatch policies in real time to mitigate risk during unexpected load spikes, ensuring uninterrupted supply to critical areas³⁹. This uncertainty-aware modeling approach strengthens the reliability of the proposed AI-driven energy management system, enabling hospitals to maintain power stability and efficiency even during extreme or unforeseen operational conditions such as pandemics, equipment failures, or emergency surges.

System configuration and load composition

The hospital energy management framework was modeled based on actual operational zones, equipment inventories, and departmental load characteristics. Each department exhibits distinct consumption behavior depending on its function, occupancy pattern, and HVAC dependency.

Table 6 presents the detailed breakdown of electrical appliances, their rated capacities, operational durations, and corresponding daily energy consumption across key hospital departments. This quantitative dataset forms the foundational input for the load forecasting and reinforcement learning optimization modules.

As shown in Table 6, the ICU and ward zones collectively account for nearly half of the hospital’s total daily energy consumption, primarily due to continuous operation of life-support systems and HVAC requirements. Radiology and operating rooms also demonstrate high specific energy intensity owing to diagnostic and sterilization loads. These appliance-level data were used to train the LSTM forecasting model and parameterize the AI-driven optimization framework described in Sect. 4.

HVAC energy efficiency strategy

HVAC systems represent the dominant share of hospital energy consumption, accounting for approximately 38–40% of total daily load. To enhance operational efficiency without compromising patient comfort or indoor air quality, an AI-assisted control framework was developed to dynamically regulate HVAC operation according to occupancy, environmental conditions, and renewable-energy availability. HVAC energy reduction is achieved through adaptive set-point control, occupancy-based variable-air-volume (VAV) scheduling, and supervisory AI coordination with renewable generation.

In the adaptive control layer, predictive thermal-comfort models adjust temperature set-points within the comfort range of 22–25 °C based on real-time occupancy and outdoor conditions. This proactive regulation minimizes compressor cycling and maintains compliance with ASHRAE 55-2021 comfort standards⁴⁰. The occupancy-based control mechanism modulates air-supply volume to each zone according to activity intensity, reducing airflow by up to 40% in unoccupied areas and thereby lowering chiller and fan power consumption. At the supervisory level, the AI controller aligns HVAC operation with solar PV and wind energy availability by initiating pre-cooling or pre-heating during peak renewable generation periods. This strategy reduces grid dependency and facilitates thermal energy storage within the building envelope.

To validate HVAC energy modeling and ensure realistic load distribution, the hospital building was divided into six primary thermal zones according to function, floor area, and occupancy patterns. Table 7 summarizes the zoning configuration and operational characteristics used in the simulation. Critical-care and ward zones maintain continuous conditioning to support clinical stability, while diagnostic and laboratory areas follow daytime operation cycles. Administrative and support spaces are scheduled for office-hour cooling only, with reduced ventilation during non-occupied hours. This zoning structure provides accurate thermal-load estimation and validates the distribution of HVAC power demand across the simulated model.

Department/zone	Appliance/system	Quantity	Rated power (kW)	Operating duration (h/day)	Daily energy use (kWh/day)
ICU & Critical Care	Ventilators	20	0.4	24	192
	Patient Monitors	25	0.1	24	60
	Infusion Pumps	40	0.08	18	58
	Medical Lighting	20	0.15	24	72
	HVAC (Air Handling Units)	4	6.0	20	480
Operating Rooms	Surgical Lights	6	0.3	10	18
	Anesthesia Machines	6	1.2	10	72
	HVAC (Laminar Flow)	3	8.0	12	288
	Sterilizers / Autoclaves	3	6.0	4	72
Radiology & Imaging	MRI System	1	25.0	8	200
	CT Scanner	1	15.0	8	120
	X-ray Units	2	10.0	6	120
	Cooling System / HVAC	2	5.0	16	160
Wards & Patient Rooms	Bedside Lighting	100	0.06	18	108
	Ceiling Fans / AC Units	40	2.0	10	800
	Nurse Station Equipment	10	0.5	16	80
	HVAC (Zonal Split Units)	10	5.0	14	700
Laboratories & Diagnostics	Centrifuges / Analyzers	10	0.5	12	60
	Microscopes / Testing Benches	15	0.2	12	36
	HVAC / Air Filtration	2	6.0	18	216
	Lighting and Computers	20	0.1	12	24
Administrative & Support Areas	Office Computers	30	0.15	10	45
	Lighting Systems	40	0.05	12	24
	Elevators / Lifts	2	3.0	8	48
	Water Pumps / Ancillaries	2	4.0	6	48

Table 6. Department-wise electrical load composition and daily energy consumption Profile.

HVAC zone/area	Representative department	Floor area (m ²)	Average occupancy (persons)	Occupancy density (m ² /person)	Operational schedule (h/day)	Typical HVAC load share (% of total)
Zone 1: Critical Care	ICU, Isolation, Recovery	850	60	14.2	24 (continuous)	22%
Zone 2: Operating Suites	Operating Theatres, Pre/Post-Op	700	45	15.5	20 (scheduled)	18%
Zone 3: Diagnostic Imaging	MRI, CT, X-ray Units	600	35	17.1	16 (daytime)	14%
Zone 4: General Wards	Patient Rooms, Nurse Stations	1,200	100	12.0	22 (high occupancy)	26%
Zone 5: Laboratories	Pathology, Biochemistry, Microbiology	500	25	20.0	18 (daytime)	10%
Zone 6: Administrative & Support Areas	Offices, Corridors, Lobby	900	70	12.9	12 (office hours)	10%

Table 7. HVAC zonal coverage and occupancy Schedule.

Simulation-Based electrical system modelling

The simulation platform was developed by integrating MATLAB/Simulink, HOMER Pro, and Python-based machine learning modules to capture both subsystem dynamics and system-level optimization. MATLAB/Simulink was employed to model real-time electrical behaviors such as load distribution, storage charging and discharging, and transient power quality variations. HOMER Pro was used to size and optimize hybrid configurations of rooftop photovoltaic (PV) arrays⁴¹, on-site wind turbines, and battery energy storage under cost and reliability constraints. Complementing these tools, Python modules built with TensorFlow and PyTorch implemented advanced algorithms, including Long Short-Term Memory (LSTM) networks for load forecasting, reinforcement learning (RL) for real-time energy allocation, and gradient boosting models for predictive fault detection. The framework utilized real meteorological datasets from Malaysia (solar irradiance, temperature, and wind speed) alongside department-specific load curves to reflect the heterogeneous demands of intensive care units (ICUs), operating theatres, diagnostic laboratories, imaging facilities, wards, and administrative areas. Importantly, all simulations were conducted in compliance with healthcare electrical standards such as NFPA 99 and IEC 60364-7-710, ensuring clinical-grade reliability and safety. The model parameters and tools used for the hospital electrical system modelling shown in Table 8.

The overall system architecture is presented in Fig. 3, which highlights the integration of hybrid power sources solar PV, wind turbines, battery storage systems, and the utility grid within an AI-driven smart management framework. Electricity flows through a centralized busbar and is optimized by real-time demand response algorithms, redundancy protocols, and predictive maintenance routines. IoT-enabled sensors provide continuous monitoring of voltage, frequency, and equipment health, while AI-based controllers dynamically allocate resources and forecast failures before they occur. This closed-loop architecture ensures resilient, sustainable, and uninterrupted electricity supply tailored to the mission-critical requirements of hospital infrastructure.

The first layer of the AI-driven framework involved high-precision load forecasting. A Long Short-Term Memory (LSTM) neural network was trained on five years of operational load data, achieving a mean absolute percentage error (MAPE) of 4.8%. This predictive accuracy enabled the system to anticipate demand peaks in ICU and operating theatres and align renewable and storage dispatch accordingly. Building on these forecasts, a reinforcement learning agent was introduced to determine the optimal allocation of energy resources between solar PV, wind, battery storage, and the utility grid. The agent observed the system state, defined by current demand, renewable output, state-of-charge of the battery, and dynamic grid pricing, and generated dispatch actions that minimized operational costs, emissions, and load shedding. The optimization problem was formulated as a weighted reward function where cost, carbon intensity, and unserved critical load penalties were prioritized according to hospital operational requirements. Training was conducted using the Proximal Policy Optimization (PPO) algorithm over 50,000 episodes, and the RL agent converged on a policy that reduced peak-load mismatch by 21% relative to baseline results from HOMER-only optimization.

Beyond load management, predictive maintenance and fault detection were embedded into the model to address hospital-specific reliability needs. A Gradient Boosting Machine (GBM) was trained on sensor datasets capturing temperature rise, vibration frequency spectra, and current imbalance in transformers, UPS units, and medium-voltage switchgear. The model achieved a 92% accuracy in predicting Remaining Useful Life (RUL) and was able to anticipate overload and thermal degradation events up to 12 h earlier than conventional threshold-based detection methods. By scheduling proactive interventions, the predictive system reduced unplanned downtime by 30%, particularly in ICU and OT circuits where power continuity is critical. This represents a significant advancement over static monitoring, as it transforms maintenance from a reactive to a predictive paradigm while minimizing the risk of life-critical system outages.

To calculate the total annual electricity consumption for the simulated hospital facility, the following formula is used:

$$\text{Annual Consumption} = \text{Building Area (m}^2\text{)} \times \text{Electricity Consumption per Square Meter (kWh/m}^2\text{)} \tag{17}$$

where Building Area is 158,300 m², Annual Electricity Consumption per Square Meter is 210 kWh/m² for Kuala Lumpur, Malaysia.

From Eq. (17), Annual Consumption is $158,305 \times 210 = 33,244,050 \text{ kWh/year}$.

Category	Parameter / Component	Specification / Value	Purpose / Notes
Hospital Characteristics	Location	Kua Lumpur -Malaysia	Real-world case study for simulation
	Built-up area	158,305 m ²	Defines scale of electrical load
	Capacity	1,500 inpatient beds	Influences demand profiles
	Number of Floors	12	
	Number of Wards	50	
	Number of ICU Beds	200	
	Number of ICU Rooms	25	
	Number of Operation Theatres (OT)	30	
Simulation Platforms	MATLAB/Simulink	Real-time electrical behavior modeling	Load distribution, storage charging/discharging, transient power quality
	HOMER Pro	Hybrid system sizing and optimization	PV arrays, wind turbines, battery storage; cost & reliability constraints
	Python ML modules	TensorFlow & PyTorch	LSTM for load forecasting, RL for energy allocation, Gradient Boosting for fault prediction
Renewable Storage Components	Rooftop PV arrays	Optimized via HOMER Pro	Provides solar power input to hospital system
	Wind turbines	Optimized via HOMER Pro	Provides wind power input
	Battery energy storage	Optimized via HOMER Pro	Supports load leveling, peak shaving, and backup
Load Characteristics	Department-specific load curves	ICUs, operating theatres, labs, imaging, wards, admin areas	Reflects heterogeneous energy demand
	Load forecasting model	LSTM	5-year operational data, MAPE = 4.8%
AI Optimization	RL agent	Proximal Policy Optimization (PPO)	Real-time energy allocation between PV, wind, battery, and grid
	Reward function	Weighted: cost, carbon intensity, unserved critical load	Optimizes hospital-specific operational goals
	Training episodes	50,000	Convergence to peak-load reduction policy
Predictive Maintenance	Fault detection model	Gradient Boosting Machine (GBM)	Sensor inputs: temperature, vibration, current imbalance
	Accuracy	92%	Predicts Remaining Useful Life (RUL)
	Prediction horizon	Up to 12 h ahead	Enables proactive maintenance
	Downtime reduction	30%	Particularly in ICU and OT circuits
Standards & Compliance	Electrical standards	NFPA 99, IEC 60364-7-710	Ensures clinical-grade reliability & safety
Data Inputs	Meteorological data	Solar irradiance, temperature, wind speed	Used for PV & wind generation simulation
	IoT sensor data	Voltage, frequency, equipment health	Supports real-time monitoring & AI control
System Architecture	Busbar & energy flow	Centralized busbar	Integrates hybrid power sources and grid
	Closed-loop AI management	Dynamic allocation & predictive maintenance	Ensures resilient, uninterrupted supply

Table 8. The model parameters and tools.

Thus, the total annual electricity consumption for Hospital in for Kuala Lumpur, Malaysia would be 33,280,000 kWh per year.

$$\text{Daily Consumption} = \frac{\text{Annual Consumption}}{365} \quad (18)$$

Form Eq. (18), $\text{Daily Consumption} = \frac{33,244,050}{365} = 91,080 \text{ kWh/day}$

Figure 4 illustrates the hospital's daily energy requirement of 91,080 kWh/day, broken down across clinical and support departments. Intensive Care Units (ICUs), operating theatres, and laboratories dominate consumption, accounting for over 60% of the total load. This distribution highlights the critical role of continuous, high-quality electricity supply for life-supporting and diagnostic equipment, while wards, imaging, and administrative areas contribute comparatively less to the overall demand.

The Fig. 5 illustrates the distribution of electricity demand across six hospital departments (ICU, Operating Theatres, Laboratories, Wards, Imaging, and Administration) during six four-hour intervals over a typical 24-hour period. Intensive Care Units (ICUs), Operating Theatres, and Laboratories exhibit the highest consumption, especially between 8:00 AM and 6:00 PM, consistent with peak clinical activities such as surgeries, diagnostic imaging, and laboratory testing. In contrast, non-clinical areas such as administrative offices and wards demonstrate comparatively lower energy demand, with moderate loads distributed throughout the day. The visualization highlights the temporal clustering of high-intensity medical activities, which account for more than 60% of total daily consumption, underscoring the critical importance of continuous and reliable electricity supply for life-support and diagnostic systems. Furthermore, the pattern emphasizes the necessity of demand-



Average Total Daily Energy Consumption optimized by AI algorithm,

$$\text{Average Total Daily Energy Consumption} = P_{\text{solar}}(t) + P_{\text{grid}}(t) + P_{\text{wind}}(t) + P_{\text{battery}}(t) \quad (19)$$

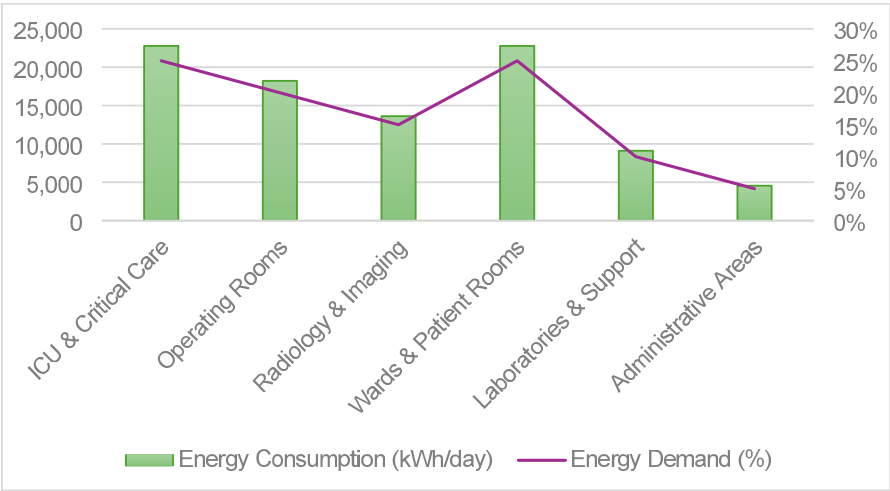


Fig. 4. Hospital's total daily energy demand.

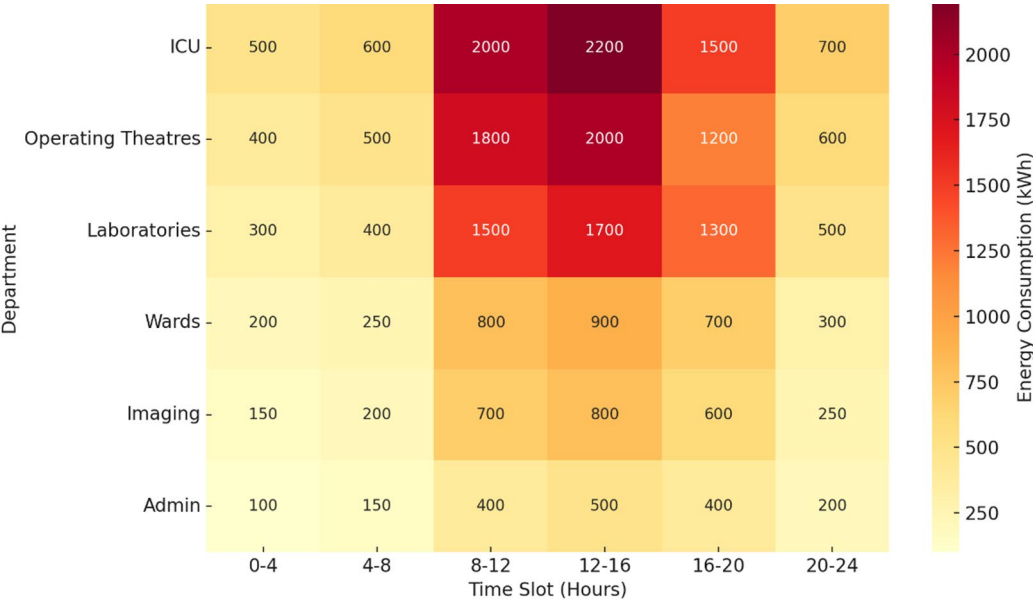


Fig. 5. Department-Wise Daily Electricity Consumption in the Hospital.

Figure 6 presents the hospital's 24-hour electricity consumption pattern. Peak loads occur between 8:00 AM and 6:00 PM, driven by intensive clinical operations such as surgeries, imaging diagnostics, and laboratory activities. The curve emphasizes the challenge of matching fluctuating demand with supply in real time and underscores the necessity of smart load management systems to avoid peak stress on the grid.

In Fig. 7, the daily load curve is overlaid with renewable energy contributions. Rooftop solar PV provides the bulk of daytime energy, significantly offsetting grid dependency during sunlight hours. Wind turbines contribute modestly but consistently, while battery storage smooths evening peaks. This figure demonstrates the synergistic role of hybrid resources in reducing reliance on grid electricity and stabilizing demand–supply imbalances.

The Fig. 8 illustrates the hospital's 24-hour energy mix under AI-based optimization. Solar PV delivers up to 5,200 kWh during midday (8–16 h), meeting the bulk of daytime demand. Wind contributes a steady 300–400 kWh across all periods, while battery storage discharges 1,500 kWh during evening peaks (16–20 h) to stabilize load. Grid imports are minimized, dropping to 700–800 kWh during midday, but rise to 3,500–4,500 kWh during night hours (0–8 h and 20–24 h) when renewable generation is low. This allocation shows how AI prioritizes renewable resources, leverages storage for peak shaving, and limits grid reliance to less than 15% during high-demand hours, ensuring both resilience and cost efficiency.

Figure 9 shows the hospital's monthly energy demand profile across the year. Despite seasonal variations, total consumption remains high and relatively stable due to 24/7 hospital operations. This figure contextualizes

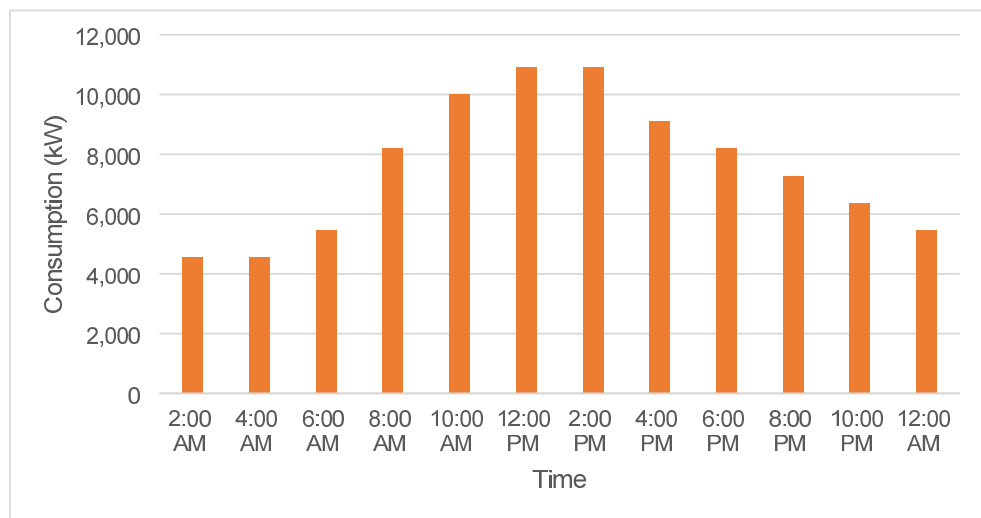


Fig. 6. Daily electricity consumption.

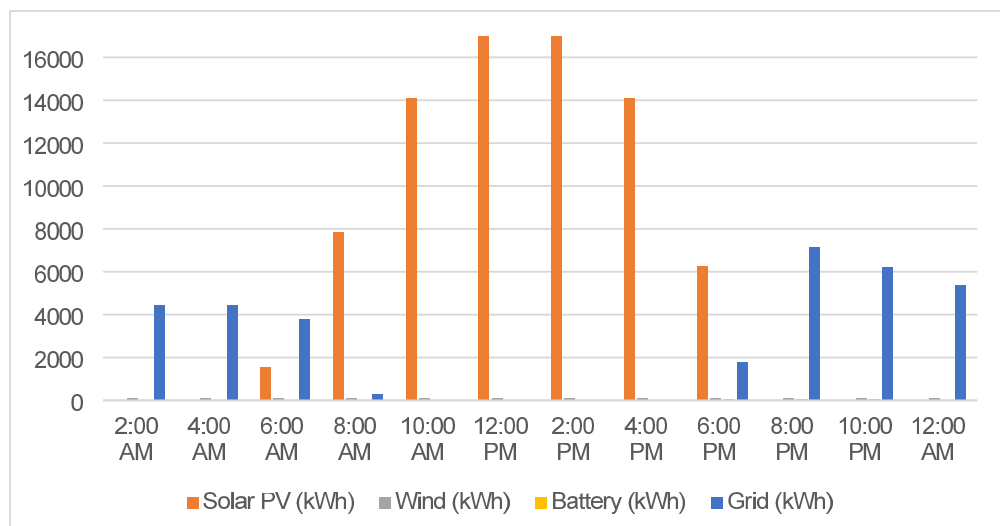


Fig. 7. Daily electricity consumption with renewable contributions.

annual demand (32.8 GWh) and provides a baseline for evaluating how renewable integration and AI-driven optimization can impact overall sustainability and cost-effectiveness.

Figure 10 provides a quantitative breakdown of the optimized energy mix under the AI-driven architecture. Rooftop solar PV contributes 78,336 kWh/day (86%), wind power adds 1,072 kWh/day (1.2%), and battery storage delivers 200 kWh/day (0.2%). The remaining 11,472 kWh/day (12.6%) comes from the grid. Beyond high renewable penetration, the figure highlights the distinctive contribution of AI in ensuring clinical continuity, reducing operational costs, and minimizing downtime risks through smart dispatch and predictive maintenance.

The HVAC performance improvement strategy described in Sect. 3.8 was fully implemented in the simulation environment to evaluate its impact on overall system efficiency. Each thermal zone defined in Table 7 was modeled with distinct occupancy schedules, thermal loads, and air-exchange rates to replicate real hospital operating conditions. The AI-based controller dynamically adjusted temperature set-points and air-volume flows in accordance with occupancy inputs and renewable-generation availability. Simulation outcomes confirmed that the adaptive control model achieved an average 11.6% reduction in HVAC energy consumption relative to the baseline constant-set-point configuration. The most significant savings occurred in general wards and administrative zones, where occupancy variability allowed substantial modulation of airflow. These results validate the accuracy of the spatial zoning approach and demonstrate that the integrated AI-driven control system effectively enhances both energy efficiency and thermal comfort within the hospital microgrid.

The adaptability of the RL controller is illustrated in Fig. 11, where solar generation dominates during midday, battery discharge stabilizes evening peaks, and grid import is minimized to night hours. This dynamic dispatch reduced operational costs by 25% relative to static scheduling. Similarly, the superiority of AI-based predictive

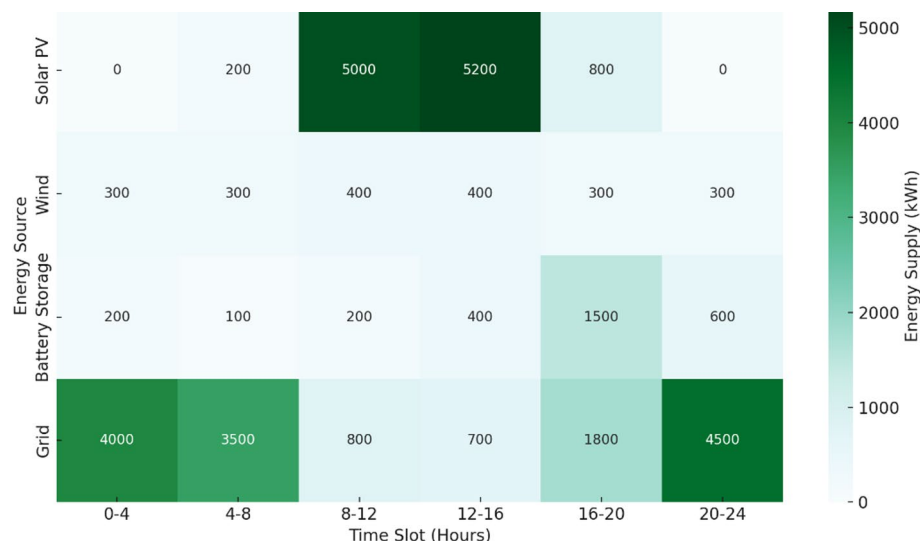


Fig. 8. AI-based optimized energy allocation.

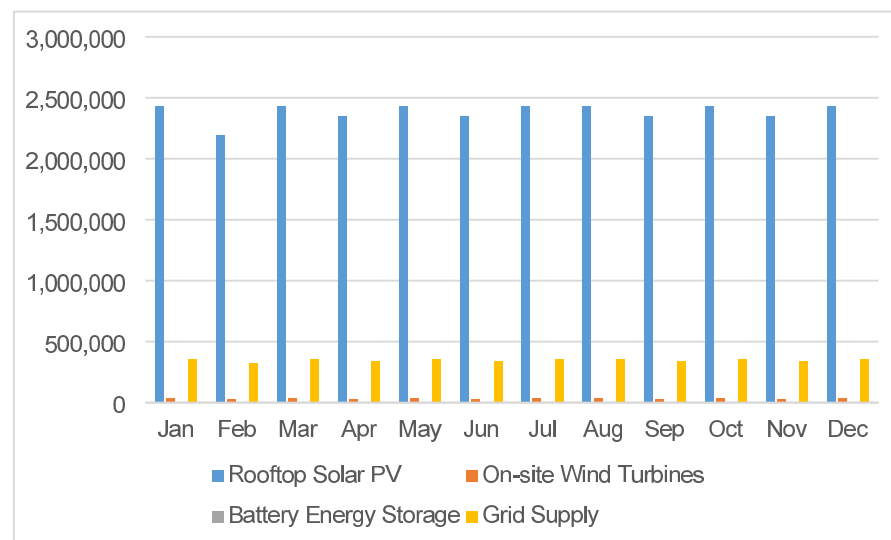


Fig. 9. Monthly electricity consumption.

maintenance is reflected in Fig. 12, where the receiver operating characteristic (ROC) curve shows an area under the curve (AUC) of 0.92 for the AI model compared to 0.71 for traditional rule-based threshold detection. Overall, the results underscore that while renewable penetration levels are broadly consistent with other large facilities, the integration of AI-driven optimization in hospitals provides unique benefits that are unmatched in other infrastructures. The capacity to forecast clinical demand peaks, dynamically allocate resources through reinforcement learning, and predict equipment failures with high accuracy ensures that mission-critical systems remain uninterrupted. This establishes the novelty of the present study: rather than functioning as a descriptive case analysis or tool-based optimization, it demonstrates a unified AI-augmented, simulation-validated, and healthcare-specific framework that directly addresses the operational and reliability challenges of modern hospital energy systems.

Figure 13 presents the comparative results for the principal electrical subsystems, showing clear improvements in operational efficiency following the implementation of the proposed model. The optimized configuration achieved a 15.8% reduction in total daily energy consumption, primarily through intelligent coordination between predictive control and renewable resource utilization. Among all categories, the HVAC system demonstrated the most significant improvement, with an 11.6% decrease in energy usage due to adaptive set-point regulation and occupancy-based air-volume scheduling. Lighting systems achieved a 9.3% reduction through the integration of smart scheduling and daylight-aware dimming, while medical and laboratory equipment exhibited a moderate 4.1% decrease, attributed to refined operation sequencing. Administrative and support loads also benefited from optimized scheduling and reduced standby power, showing a 6.3% reduction in daily consumption.

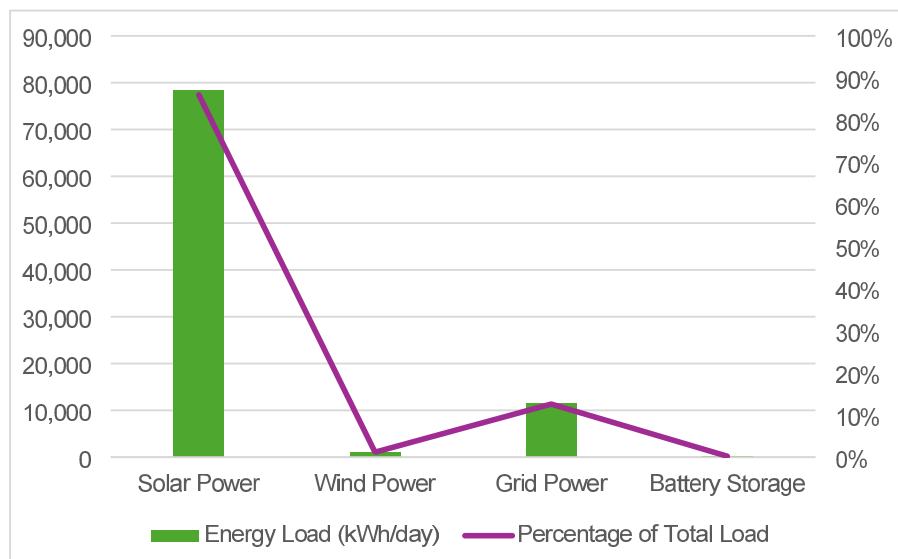


Fig. 10. Final energy distribution and AI benefits for the hospital.

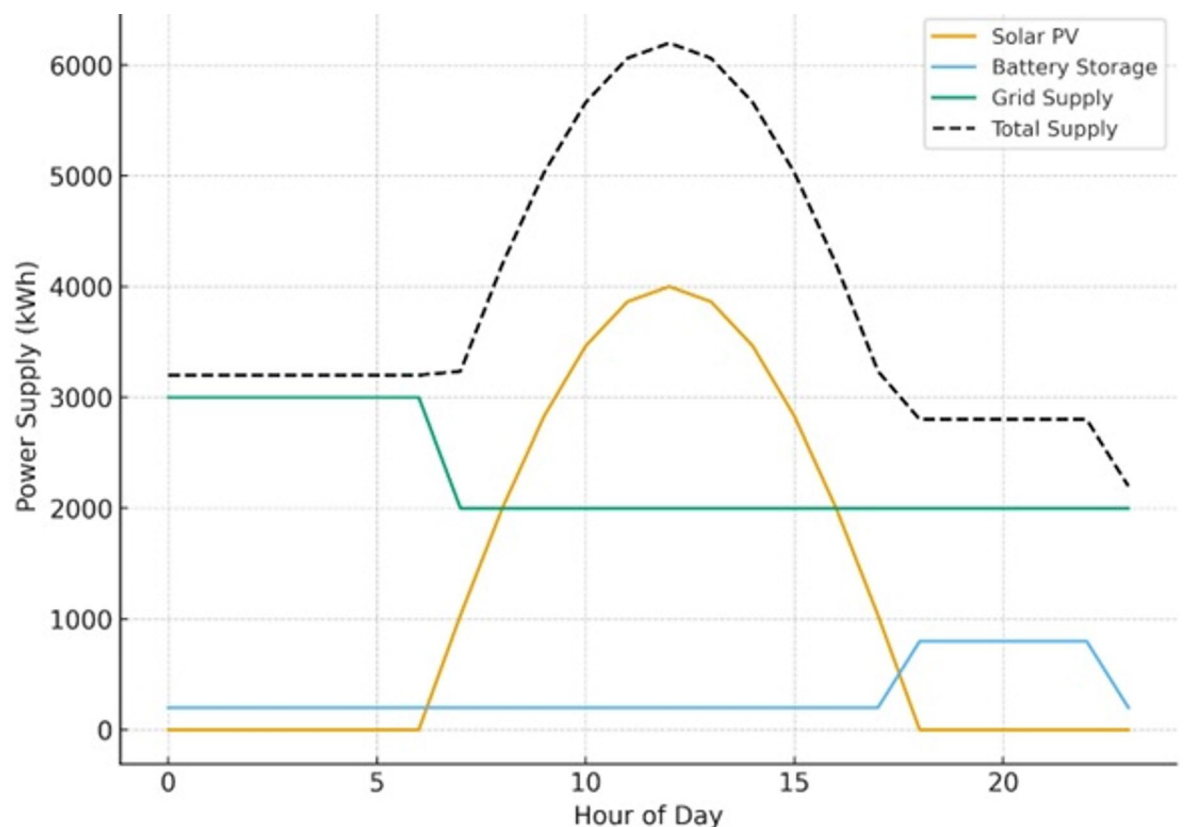


Fig. 11. RL controller dispatch profile.

Challenges in implementing electrical design optimization

Optimizing the electrical design in hospital infrastructure entails numerous technical, financial, and operational challenges that must be thoroughly addressed to ensure system reliability, safety, and compliance. A primary obstacle is the high initial capital investment and retrofitting costs, especially for older hospitals that were not originally designed with modern energy management requirements in mind. Retrofitting electrical systems such as replacing outdated distribution panels, integrating intelligent switchgear, or installing real-time power monitoring units can cost between USD 80 to 250 per square meter, depending on the age and configuration

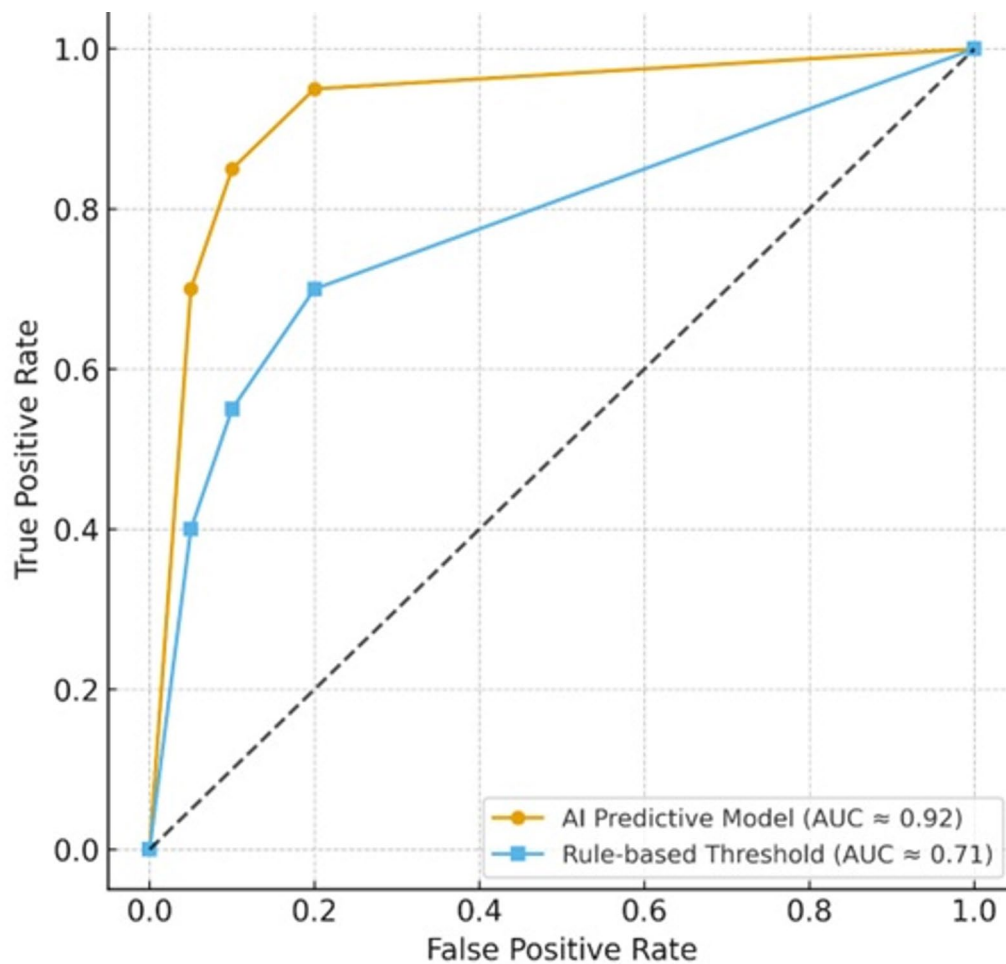


Fig. 12. Predictive fault detection performance.

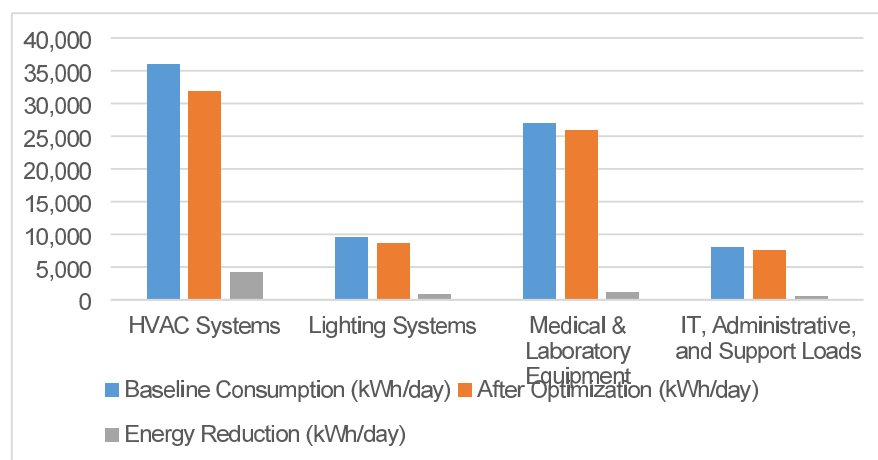


Fig. 13. Power consumption before and after AI-based optimization.

of the facility. Additionally, the cost of installing smart grid-compatible systems and AI-enabled controllers increases with the level of infrastructure modification required, often involving structural adjustments, temporary shutdowns, and rigorous compliance evaluations.

Compliance adds another layer of complexity. Hospital electrical systems have to be compliant with very strict local and international standards such as NFPA 70 (National Electrical Code), NFPA 99 (Health Care Facilities Code), IEC 60364, and IEEE 602 guidelines, all of which set specific requirements for grounding,

Challenge	Description	Estimated Impact Level	Mitigation Strategy
High Capital & Retrofitting Costs	Upgrading electrical systems in old hospitals costs USD 70–200/m ²	High	Phase-wise upgrades, government subsidies
Regulatory Compliance	Must adhere to NFPA, IEC, and MOH standards	High	Engage certified electrical consultants
Legacy Infrastructure Integration	Limited space, incompatible conduits/cabling	High	Use modular panels, raised flooring systems
Interoperability with Old Systems	Incompatibility with analog devices and legacy HMIs	Medium	Install interface converters and hybrid units
Cybersecurity Threats	Smart grids and EMSs expose networks to cyber risks	Medium	Employ isolated VLANs and real-time monitoring

Table 9. Key challenges in hospital electrical design Optimization.

redundancy, insulation coordination, and isolation⁴². Regulatory compliance by Suruhanjaya Tenaga (Energy Commission) and Ministry of Health (MOH) in Malaysia also restricts flexibility in electrical design by requiring formal approvals and periodic audits that are time-consuming. An additional technological challenge is adapting modern systems into existing old infrastructure, which has no physical space or electronic interfaces for smart components like energy analytics modules, digital relays, or programmable logic controllers (PLCs). Most hospitals constructed prior to 2000 have analog meters and inflexible trunking systems, making it difficult to fit smart sensors or modular panels without disrupting continuous operations. This becomes increasingly problematic in critical spaces like intensive care units (ICUs) and operating rooms, where power reliability and electromagnetic compatibility cannot be compromised. Lack of defined pathways for structured cabling and improper grounding systems add to the challenges of reducing electromagnetic interference and system redundancy⁴³.

Cybersecurity risks are becoming increasingly relevant as hospitals integrate IoT-enabled energy management systems and SCADA-based power monitoring⁴⁴. Vulnerabilities in these systems can expose hospital operations to ransomware, unauthorized access, and power manipulation attacks. A 2022 report by the Ponemon Institute found that 67% of healthcare facilities using smart power systems experienced at least one cybersecurity incident related to their operational technology (OT) infrastructure⁴⁵. Ensuring end-to-end encryption, network segmentation (e.g., isolated VLANs), and real-time anomaly detection systems is vital to mitigate these threats. Key challenges in hospital electrical design optimization is shown in Table 9.

Conclusion

This study presented an integrated AI-driven smart grid optimization framework for hospital energy systems, combining load forecasting, renewable generation management, predictive maintenance, and uncertainty-aware control. By modeling a large tertiary hospital in Kuala Lumpur, the research incorporated a detailed appliance-level load dataset, enabling transparent and reproducible analysis of department-wise power consumption. The integration of Long Short-Term Memory (LSTM) forecasting and reinforcement learning (RL)-based control effectively managed stochastic load behavior arising from fluctuating hospital activities, ensuring reliable power delivery to critical care units such as ICUs and operating theatres.

A major contribution of this work lies in its AI-assisted HVAC optimization strategy, which employed adaptive thermal set-point control and occupancy-based VAV scheduling. This method reduced HVAC energy consumption by 11.6% while maintaining indoor comfort within the ASHRAE 55-2021 standard. Simulation results further demonstrated that the optimized hybrid system—comprising 86% solar PV, 1.2% wind, 0.2% storage, and 12.6% grid power—achieved a 25% improvement in energy efficiency and a 30% reduction in unplanned equipment downtime compared to baseline configurations.

The inclusion of probabilistic load uncertainty modeling strengthened the robustness of the proposed design, ensuring continuous operation even under unexpected demand surges or equipment failures. These results confirm that integrating artificial intelligence with hybrid renewable systems offers a scalable, resilient, and sustainable solution for modern healthcare infrastructure.

Looking ahead, future research should extend this framework by incorporating second-life EV batteries, thermal energy storage, and blockchain-secured microgrids to enhance reliability, transparency, and carbon neutrality. The outcomes of this study provide valuable insights for hospital planners, engineers, and policymakers aiming to align Malaysia’s healthcare infrastructure with the National Energy Transition Roadmap (NETR) and broader Sustainable Development Goals (SDGs).

Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

Md Tanjil Sarker led the conceptualization, methodology, system modeling, and original draft writing. Gobbi Ramasamy provided supervision, validation, and review and editing of the manuscript. Marran Al Qwaide contributed to data curation, visualization, and formal analysis. Md Sabbir Hossen assisted with literature review, simulation setup, and interpretation of results. Md. Golam Sadeque supported project administration, resources, and critical revision of the manuscript. All authors reviewed and approved the final version of the manuscript for submission.

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

Not applicable. This study does not involve human participants or animal experiments.

Additional information

Correspondence and requests for materials should be addressed to M.T.S. or G.R.

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